

# cvbets: a fully autonomous esports prediction system

cvbets is a simulated prediction-tracking system for professional Counter-Strike 2 matches — built to answer one question with real data: can AI agents learn to predict match winners, and can they be honest about how confident they are? Four AI agents compete nightly. An append-only ledger keeps the receipts. A public dashboard shows the scoreboard. Everything runs in the cloud, 24/7, with no human in the loop and no real money anywhere.

## The architecture — four computers, zero laptops

| Component          | Runs on                        | Job                                                                                                                                                                      |
|--------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nightly pipeline   | GitHub Actions (free)          | 07:00 UTC daily: ingest matches from PandaScore, chalkbot/form/h2h submit picks, debate fires, grading settles results, dashboard rebuilds, ledger committed back to git |
| sage (LLM expert)  | Hetzner VM, Germany (~\$14/mo) | 07:45 UTC daily: reads the ledger + debate, reasons with an LLM (GLM via Ollama cloud), submits its own pick through the same gate                                       |
| Conversational bot | Same Hetzner VM                | 24/7 Discord bot — four humans authorized; asks and answers, can edit its own code                                                                                       |
| Dashboard          | GitHub Pages (free)            | Auto-refreshes on every push — six tabbed views                                                                                                                          |
| Watchdog           | GitHub Actions                 | Alerts Discord if the nightly loop ever fails                                                                                                                            |
| State backups      | Private GitHub repo            | Nightly: sessions, memory, skills, config                                                                                                                                |

The user's laptop is a window, not a component. The entire system survives the laptop being closed for good.

## The four competitors

| chalkbot | The favorite-chaser | Deterministic. Backs the team with the higher 30-day win rate in the ledger; coin-flip when data is thin. The baseline everyone must beat.                                                              |
|----------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| form     | The recency scout   | Deterministic. Only the last 5 matches matter — momentum over history.                                                                                                                                  |
| h2h      | The history buff    | Deterministic. Only the all-time head-to-head record between the two teams — matchups over streaks.                                                                                                     |
| sage     | The reasoner (LLM)  | Reads the full ledger context — both teams' form, the fixture history, every debate argument filed by the others — then reasons and commits to a pick with confidence. Powered by GLM via Ollama cloud. |
| (chief)  | Coming: the judge   | Planned: weights every expert by its proven Brier score and emits consensus picks. Unlocks when each agent has 30+ graded picks.                                                                        |

## The integrity model — why the numbers can be trusted

|                    |                                                                                                                                                                                                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Append-only ledger | Picks are written in pen. No agent may edit or delete one. The only allowed mutation: the grader stamps WIN/LOSS once, after the match. 300+ picks, zero rewrites.                                                                                                     |
| The gate           | Agents can't touch the database. Submissions go through a validating checkpoint: exact JSON contract, selection must be one of the two teams, confidence 0.5-1.0, reasoning under 100 words, one pick per agent per match. 14 rejection tests + duplicate probes pass. |
| Brier scoring      | Every graded pick gets (confidence - outcome) squared. Confidently wrong hurts far more than humbly wrong — a 90% miss costs 0.81, a 50% coin-flip miss only 0.25. Agents must be as right as they claim to score well.                                                |
| Calibration view   | The dashboard bucketizes every agent's picks by confidence and shows actual win rate per bucket — overconfidence is visible, not arguable.                                                                                                                             |
| The debate arena   | Before matches, agents cross-examine each other's picks with ledger-grounded facts (form cites last-5 momentum, h2h cites fixture history, chalkbot cites 30-day rates). 70+ critiques logged. The transcript is append-only too.                                      |
| Testing            | 38 dashboard checks + 23 smoke tests + 14 gate tests + 43 sage tests — all run on every change, throwaway databases only.                                                                                                                                              |

## What the humans see

The public dashboard ([sntvnce.github.io/cvbets](https://sntvnce.github.io/cvbets)) is a dark-themed, tabbed strategy room: Overview (standings), Picks & Pending (300 pending picks across 86 matches with LIVE badges and pick ages), Results (46 graded with scores and Brier), Teams (134 expandable team profile cards — last-10 form, 30-day rates, all-time records, sparklines), Debate (the five most contested matches with every position and objection), and Calibration (confidence vs reality). Sortable, filterable, zero-JS fallback, one self-contained page.

Discord is the social layer: a bot with full tool access lives on the VM and answers four allowed humans — standings on demand, pick explanations, even code edits pushed to GitHub from a phone. Nightly summaries and debate stages are posted automatically. A tagged welcome message explains the house rules to new crew members.

## The scoreboard today

| Agent    | Graded | W-L   | Win rate | Mean Brier |
|----------|--------|-------|----------|------------|
| chalkbot | 44     | 26-18 | 59.1%    | 0.2500     |
| form     | 1      | 0-1   | 0.0%     | 0.2500     |
| h2h      | 1      | 0-1   | 0.0%     | 0.2500     |
| sage     | 1      | 0-1   | 0.0%     | 0.2500     |

Early days: 2,588 matches tracked, 305 picks logged, 46 graded, 259 awaiting results. The real experiment — does sage's debate-informed reasoning beat deterministic formulas? — needs ~100

graded picks per agent to answer. The system will produce that data on its own.

## Engineering notes worth stealing

|                         |                                                                                                                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Idempotent everything   | Every stage can run twice with zero side effects. Compression guards stop stale API data from clobbering finished results. Grading is guarded by result IS NULL. Crashes lose nothing. |
| The repo is the product | Code + ledger + dashboard live in git. Pushing IS deploying: the nightly run executes whatever is committed. Rollback = git revert.                                                    |
| Graceful degradation    | sage skips itself cleanly where Ollama doesn't exist (GitHub Actions); the dashboard renders fully with JavaScript disabled; missing debate data renders as 'unanimous pick'.          |
| One source of truth     | The SQLite ledger is committed to git nightly by the cloud. Local copies converge; nothing diverges.                                                                                   |
| Self-hosting agent ops  | The bot's own state (memories, skills, sessions) is backed up nightly by a script the bot infrastructure runs — agents maintaining agents.                                             |

## The stack

Python 3 stdlib + httpx. SQLite (WAL). Vanilla JS. GitHub Actions + Pages. One 2-vCPU VPS. Ollama cloud (GLM). Discord webhooks + bot. Total monthly cost: about \$14 — the VPS. Everything else is free tiers. No frameworks, no build step, no vendor lock-in: the whole system is ~2,000 lines of readable Python plus one HTML file.

## What's next

Phase 5 items, in the order the data will demand them: an odds source (unlocks ROI — the number that separates accurate from profitable), the 'chief' consensus agent, automated post-mortems written into the debate table (the agents' learning loop), and a season-ending calibration report per agent. The scoreboard never sleeps.